

Microeconomics

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Saving Supply

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Transactions involving time: saving and borrowing by consumers

- Preferences
- Affordable bundles
- Consumption choices
- Effects of changes in interest rates on choices

Transactions involving time

Transactions involving time:

- Agents can move resources from one period to another by saving and borrowing
- **Principal:** amount borrowed when one person lends money to another
- **Interest:** amount of money a borrower is obliged to pay a lender, over the principal (compensation)
- **Interest Rate:** amount of interest paid on a loan during a particular period, stated as a percentage of the principal
- We call R the interest rate stated in decimal

Present value and the price of a future dollar

- In lending, money is the good and interest is the price of having that good for a given period
- If you deposit 100 dollar today and the interest rate is 10% a year, next year you get 110 dollars
- How much money you need to deposit today if you want to get M dollars next year?
 - $X = M / (1 + R)$
 - X is called the **present discounted value (PDV)** of next year M dollars

Preferences for the Timing of Consumption

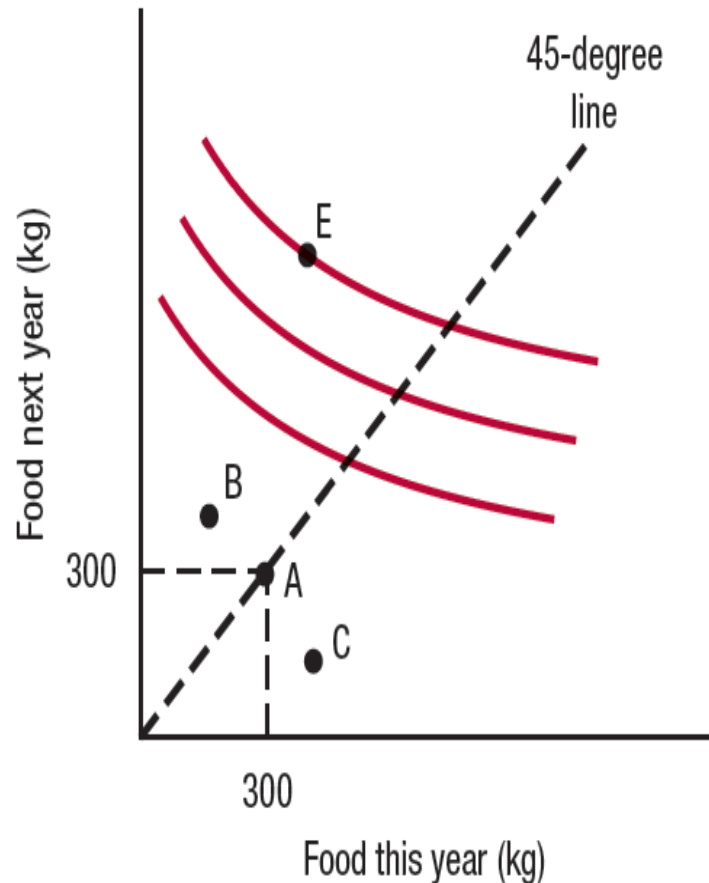
- Consider a consumer who cares about two goods: food this year and food next year
 - 2 periods t_0 and t_1
 - 2 goods c_0 and c_1
 - 2 prices p_0 and P_1
 - 2 incomes M_0 and M_1
- Agents can borrow or lend money at the interest rate R

Preferences for the Timing of Consumption

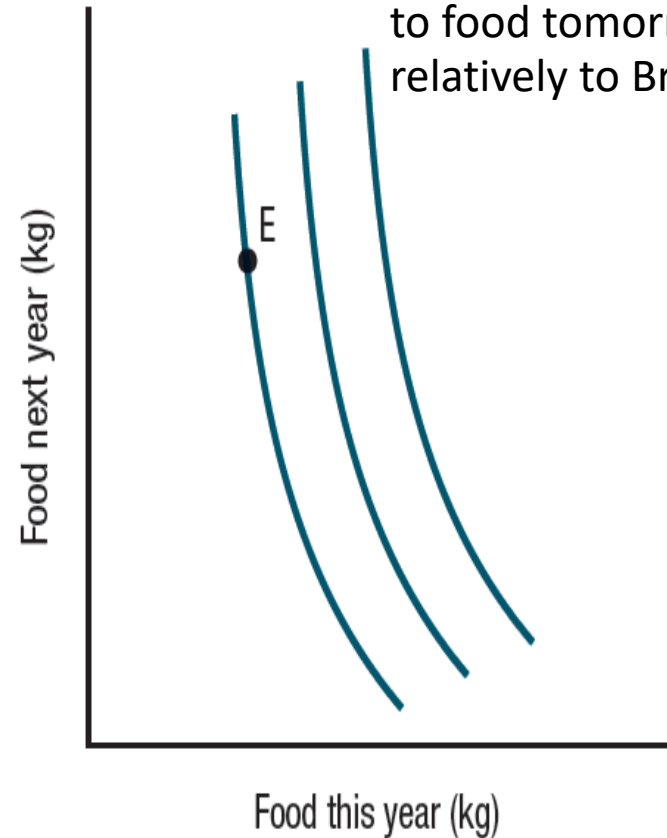
- We can use the tools we've developed so far to model consumers' decisions about the timing of consumption
- Trick: Treat identical physical objects as distinct goods if they are available at different points in time
 - $U(C_0, C_1)$

Preferences for the Timing of Consumption

(a) Brian (more patience)



(b) Ryan (less patience)



➤ Steeper indifference curves: less patience (MRS higher-Ryan prefers more food today to food tomorrow relatively to Brian)

Affordable Consumption Bundles

- A consumption bundle is affordable if, through borrowing and lending, the consumer can make all the required payments as they come due.
- Budget line:

PDV of consumption stream = PDV of income stream

$$P_0C_0 + \frac{P_1C_1}{1+R} = M_0 + \frac{M_1}{1+R}$$

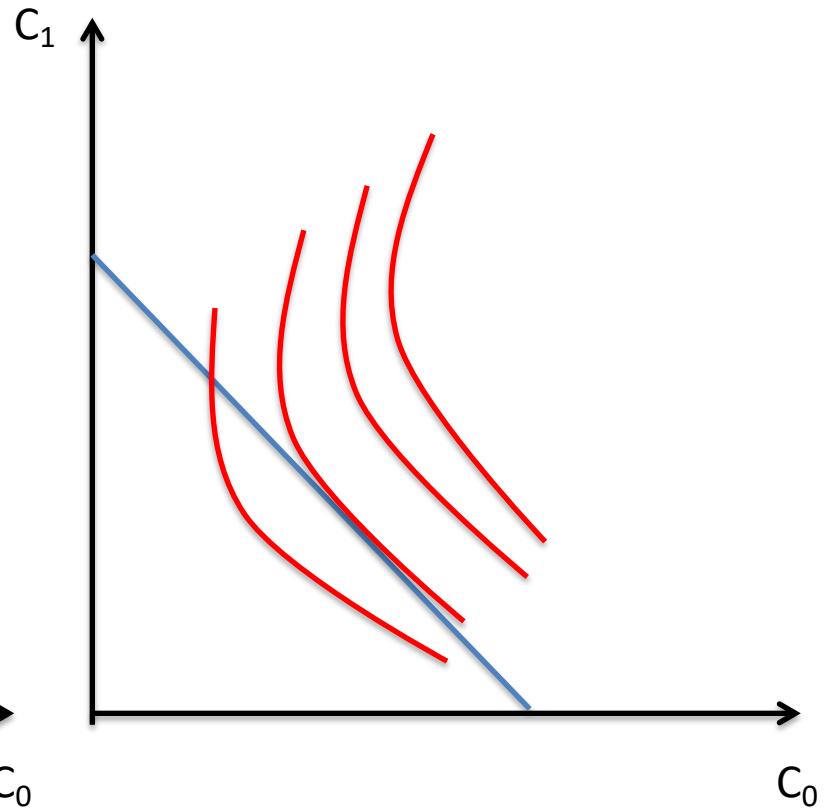
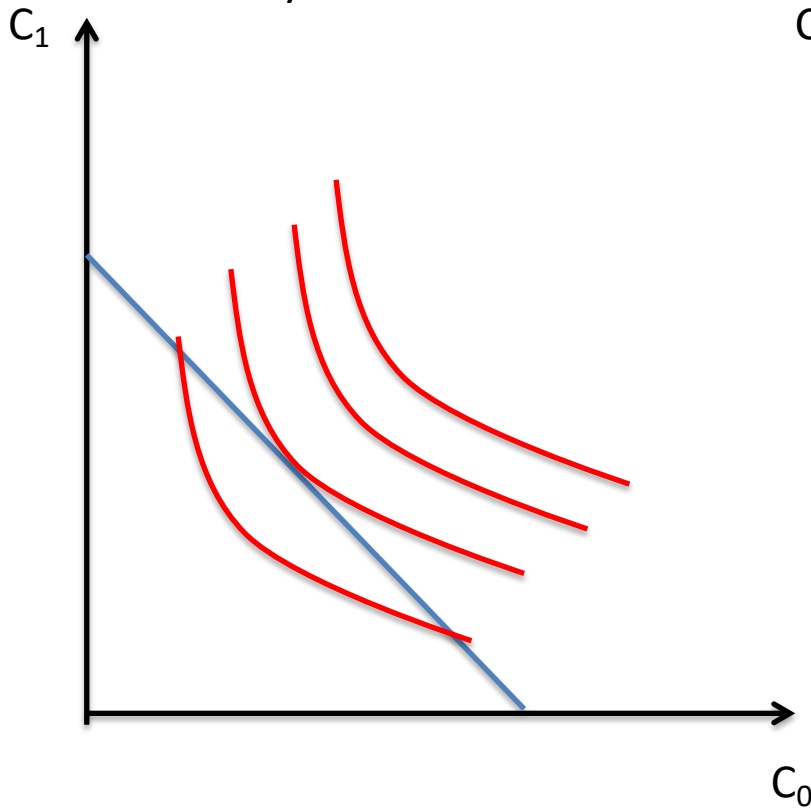
Affordable consumption bundles

- The slope of the budget is equal to the (negative) ratio of the goods' prices. P_0 is the price of the good this year and $P_1 / (1+R)$ the price of the good next year (from today's perspective).

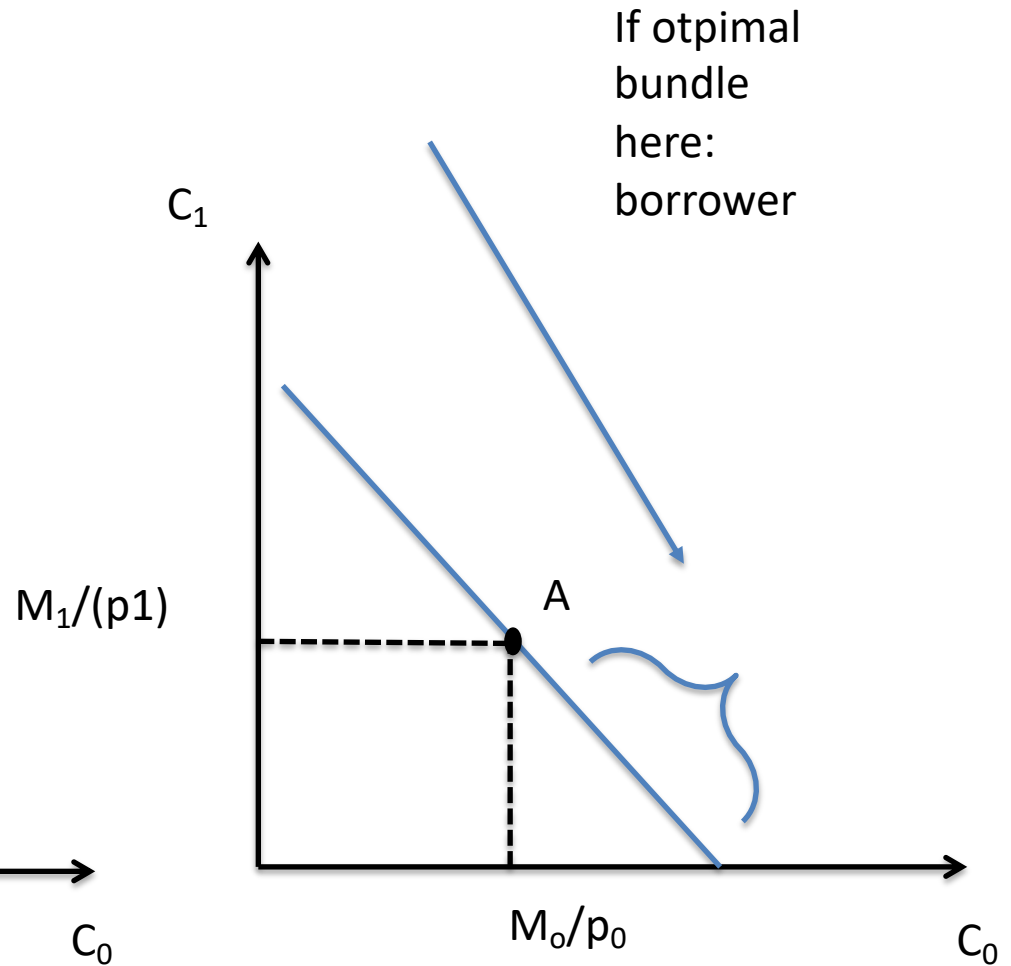
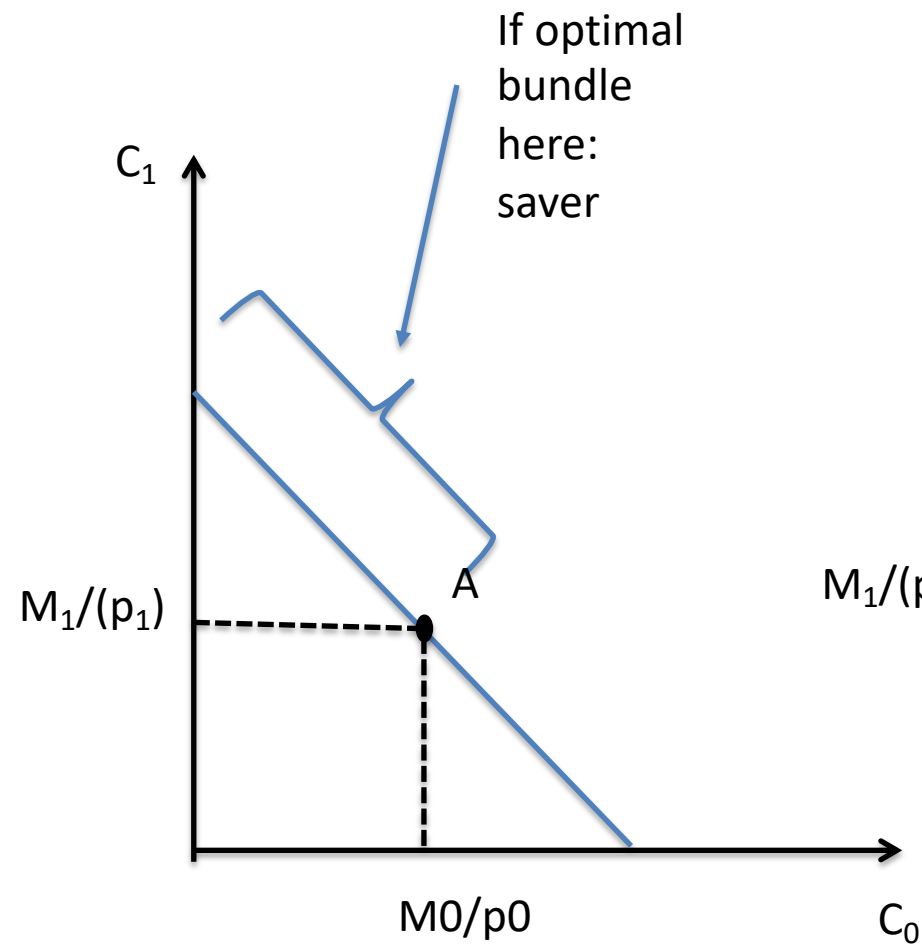
$$\text{Slope Of Budget Line} = -\frac{P_0}{P_1 / (1+R)} = -(1+R)\left(\frac{P_0}{P_1}\right)$$

Consumption Choices

- Optimal bundle: tangency condition ($MRS = \text{price ratio}$) + intertemporal budget constraint
- The more impatient you are, the more you consume today



Consumption choice

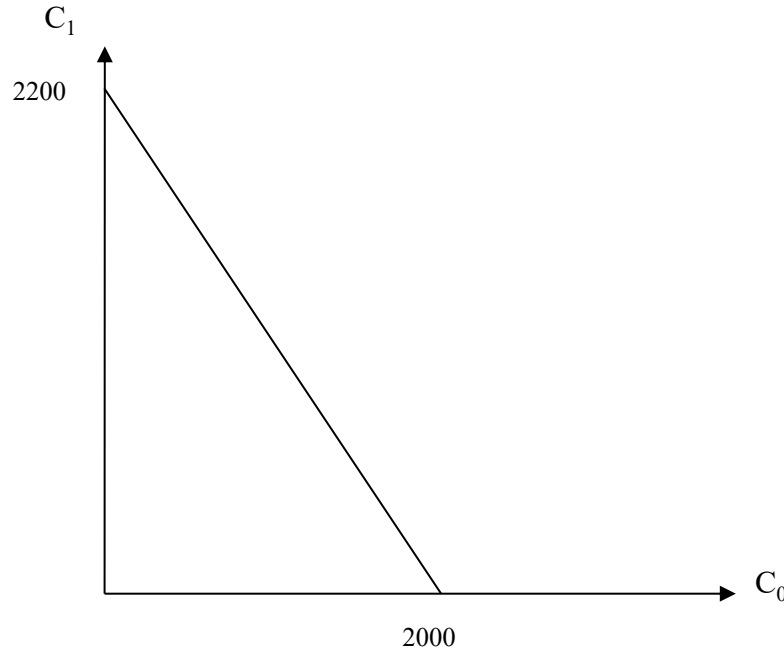


Example question

- Vittorio is hired by a firm with a contract according to which his income at time t_0 will be $M_0 = 500$ Euros, whereas at time t_1 he will get M_1 , equal to 1650 Euros. The interest rate R is equal to 0.1 and Vittorio's preferences are represented by the following utility function: $U(C_0, C_1) = C_0^{0.5} C_1^{0.5}$, where C_0 is current consumption and C_1 is future consumption. Assume price of consumption equal 1 in both periods (i.e. $P_0 = P_1 = 1$)
- Write down Vittorio's intertemporal budget constraint and represent it graphically, indicating the intercepts and the slope.

Example

- *The intertemporal budget constraint is:*
 - $C_0 + C_1/1.1 = 500 + 1650/1.1$
 - $C_1 = 2200 - 1.1C_0$
- *The intercepts are (0;2200) and (2000;0); the slope is equal to -1.1 .*



Example

- The $MRS = MU_{C_0} / MU_{C_1} = C_1 / C_0$. Derive Vittorio's optimal choice and represent it in the previous graph. Is Vittorio a saver or a borrower?

$$C_1 / C_0 = 1.1$$

$$C_1 = 1.1 C_0$$

$$1.1 C_0 + C_1 = 2200$$

We get: $C_0 = 1000$ and $C_1 = 1100$;

Vittorio is a borrower, since for him:

$$C_0 > M_0.$$

